Date: 25/04/2026

Exercise 2.1:

1. Loss: A loss is an undesirable outcome that results in harm to people, property, or the environment.

Example (Autonomous Driving):
A pedestrian being injured or killed due to a vehicle collision.

2. Hazard: A hazard is a system state or condition that, under certain circumstances, can lead to a loss.

Example (Autonomous Driving):
The vehicle continues moving while a pedestrian is present in its path.

3) Risk:
Risk is the combination of the likelihood of a hazard occurring and the severity of its consequences.

Example (Autonomous Driving):
High risk occurs when the vehicle operates in crowded urban areas where pedestrian detection may fail, increasing the chance of accidents with severe outcomes.

STPA:

Step 1(ODD, Losses, hazards)

Exercise 2.2: ODD

Dimension	Operating Condition (Valid)	Non-Operating Condition (Invalid)	How to Detect Violation
Weather	Clear or slightly cloudy weather with good visibility	Heavy rain, fog, or snow	Check visibility from camera or use rain sensor data
Lighting	Daytime with enough light	Night time or very low light	Measure brightness level from camera input
Pedestrian Type	Pedestrians similar to those seen in training data (e.g., standing or walking persons)	Pedestrians not well represented in training data (e.g., wheelchair users, unusual poses)	Use model confidence scores or detect unfamiliar patterns; improve by including diverse training data
Scene Type	Urban roads and intersections	Highways, rural areas, or off-road	Use scene classification to identify environment

Vehicle Speed	Low to moderate speed (city driving)	Very high speed driving	Monitor speed using vehicle sensors

Exercise 2.3: Unacceptable Losses:

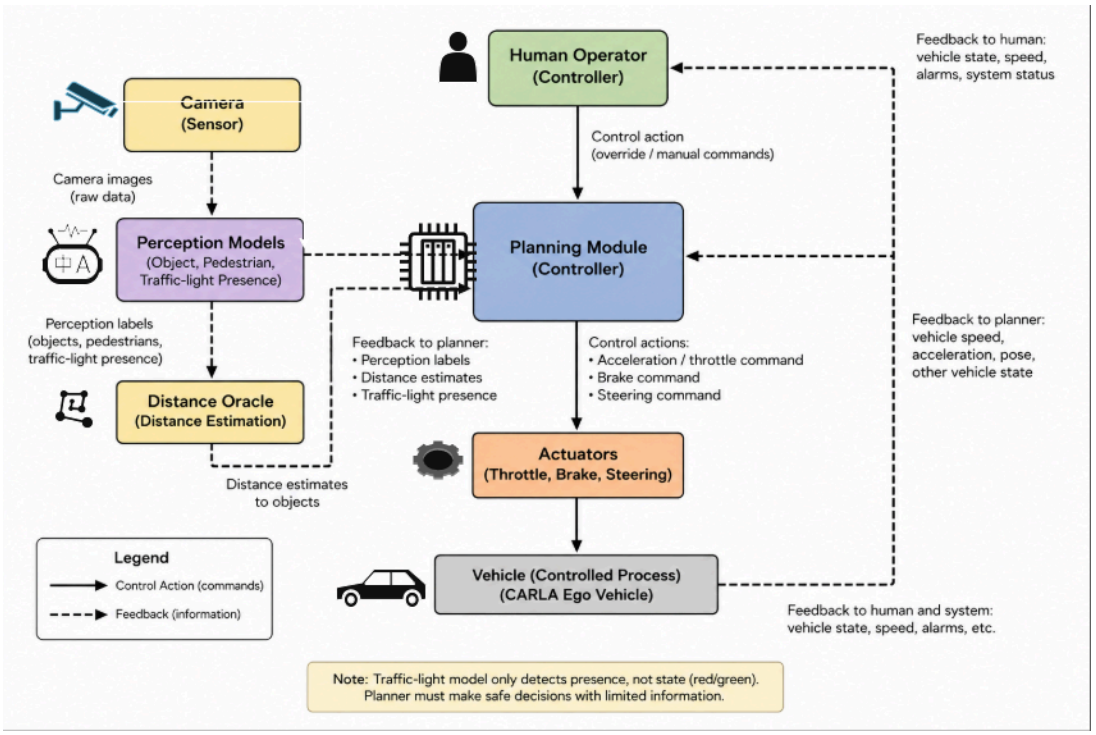
ID	Loss Description	Why Unacceptable
L-1	Human injury or death	This is the most serious outcome because it directly affects human life, which should always be protected.
L-2	Damage to the vehicle	This can lead to accidents, possible fire or dangerous situations for people inside the car.
L-3	Collision due to not obeying traffic signals	This can cause accidents, traffic jams, and can delay emergency services like ambulances, making the situation worse.

Exercise 2.4: Hazards:

ID	Hazard	Loss(es)	Likelihood	Severity
H-1	Vehicle keeps moving when a pedestrian is in front	L-1	Medium	High
H-2	System fails to detect an obstacle (pedestrian/vehicle)	L-1, L-2	Medium	High
H-3	Vehicle does not obey traffic signals (e.g., runs a red light)	L-3, L-1	Low	High
H-4	System operates outside its ODD (e.g., night or bad weather)	L-1, L-2, L-3	Medium	High

STPA Step 2: Model the Control Structure

Exercise 2.5: Control Structure Diagram



STPA Step 3: Identify Unsafe Control Actions and Derive Safety Constraints

Exercise 2.6: Unsafe Control Actions (UCAs)

ID	Controller	Control Action	UCA Type	Hazard(s)	Unsafe Scenario
UCA-1	Planning Module	Brake	Not provided	H-1	Vehicle does not brake when a pedestrian is in front
UCA-2	Planning Module	Brake	Provided unsafely	H-2	Vehicle brakes even when there is no obstacle
UCA-3	Planning Module	Brake	Wrong timing	H-1	Vehicle brakes too late after detecting a pedestrian
UCA-4	Planning Module	Brake	Wrong duration	H-1	Vehicle stops braking too early while pedestrian is still present
UCA-5	Human Operator	Take control	Not provided	H-4	Human does not take control when system is outside ODD
UCA-6	Human Operator	Take control	Provided unsafely	H-2	Human interferes unnecessarily when system is working correctly

UCA-7	Human Operator	Take control	Wrong timing	H-1	Human reacts too late to avoid a collision
UCA-8	Human Operator	Take control	Wrong duration	H-3	Human takes control briefly and releases it too early

Exercise 2.7: Safety Constraints :

UCA	Safety Constraint	Level	Verification
UCA-1	The system must apply braking when a pedestrian is detected in front	System-level	Test in simulation scenarios with pedestrians
UCA-2	The system should not brake unless a valid obstacle is detected	Model-level	Evaluate false positive rate of perception model
UCA-3	The system must respond within a safe time after detecting an obstacle	System-level	Measure response latency in tests
UCA-4	The system must continue braking until the path is clear	System-level	Simulation testing with moving pedestrians
UCA-5	The human operator must take control when the system operates outside ODD	System-level	Check system alerts and human response
UCA-6	The system should avoid unnecessary human intervention during normal operation	System-level	Monitor system behavior in safe conditions
UCA-7	The human operator should react within a safe time when intervention is required	System-level	Measure reaction time in test scenarios
UCA-8	The human operator must maintain control until the situation is safe	System-level	Evaluate behavior during takeover scenarios

STPA Step 4: Identify Causal Loss Scenarios

Exercise 2.8: Causal Loss Scenarios

UCA	Causal Scenario	Root Cause	Related Constraint
UCA-1	Camera fails to detect a pedestrian, so the system thinks the road is clear and does not brake	Poor detection or unclear image	System must brake when pedestrian is detected
UCA-2	System detects a false obstacle and brakes unnecessarily	Noise or incorrect model prediction	System should not brake without valid obstacle
UCA-3	Pedestrian is detected late, so braking happens too late	Processing delay or slow response	System must respond within safe time
UCA-4	System starts braking but stops too early while pedestrian is still present	Incorrect control logic	System must continue braking until safe
UCA-5	System operates outside ODD, but human does not take control	Human not attentive or no alert	Human must take control outside ODD
UCA-6	Human takes control even when not needed and causes disturbance	Lack of trust or confusion	Avoid unnecessary human intervention
UCA-7	Human reacts too late to take control in a risky situation	Slow reaction or distraction	Human must respond within safe time
UCA-8	Human takes control but releases it too early before situation is safe	Misjudgment of situation	Human must maintain control until safe